

Spring Boot

Prerequisite : Very good working Knowledge on Java / Application Development

Basics of Java programming language [2 Hours]

- Variables, data types, and operatorsControl structures: if-else, switch-case, loops
- Object-oriented programming (OOP) concepts: classes, objects, inheritance, encapsulation, polymorphism, constructors, and methods
- Exception handling: Understanding exceptions and error handling, try-catch blocks, Handling checked and unchecked exceptions, Finally block and resource cleanup
- Java I/O (Input/Output) Basics: Introduction to Java I/O streams, Reading and writing text files, Working with file input/output streams, File and directory manipulation using java.io package

Lambda in Java[1 Hour]

- Lambda expressions: syntax and examples
- Functional interfaces and the `@FunctionalInterface` annotation
- Default and static methods in interfaces

Introduction to Spring Framework [1 Hour]

- What is the Spring Framework?
- Core features and benefits
- Inversion of Control (IoC) and Dependency Injection (DI)
- Spring modules: Core, Beans, Context

Spring Core Concepts [1 Hour]

- Understanding the Spring container
- Configuring Spring beans using XML and annotations
- Dependency Injection (DI) patterns: constructor injection, setter injection
- Bean lifecycle: initialization and destruction

Working with Spring Annotations [1 Hour]

- `@Component`, `@Service`, `@Repository`, `@Controller` `@Autowired` and `@Qualifier` for dependency injection
- Stereotype annotations: `@Configuration`, `@ComponentScan` `@Scope` and `@Lazy` annotations

Introduction to Spring Boot and RESTful Services [1 Hour]

- Overview of Spring and Spring Boot:
- Introduction to the Spring Framework.
- Advantages of using Spring Boot for rapid application development.
- Understanding the role of Spring Boot Starters and auto-configuration.

Setting Up the Development Environment [0.5 Hours]

- Installing Java, Maven/Gradle, and an IDE optimized for Spring Boot development.
- Generating a Spring Boot project with Spring Initializer.

Building Your First REST API [1.5 Hours]

- Creating a simple RESTful service with Spring Boot.
- Discussing REST principles and the importance of stateless services.
- Mapping HTTP operations to CRUD operations in a database.

Configurations and Profiles [1 Hour]

- Managing application configurations with ``application.properties``/YAML.
- Using profiles to handle different deployment environments.
- Spring Data JPA, Service Layer, and Error Handling

Data Access with Spring Data JPA [1.5 Hours]

- Integrating a relational database with Spring Boot using Spring Data JPA.
- Creating entities, repositories, and services to handle data operations.

Business Logic and Service Layer [1 Hour]

- Structuring the service layer for clean separation of concerns.
- Applying the DTO pattern for efficient data transfer and exposure.

Error Handling and API Validation [1.5 Hour]

- Implementing global exception handling with ``@ControllerAdvice``.
- Validating request data using Spring's validation API.

Introduction to Containerization [1 Hour]

- Basics of Docker and containerization.
- Containerizing a Spring Boot application.
- Best practices for Dockerfile creation and multi-stage builds for Spring Boot.

Container Orchestration [1 Hours]

- Fine-tuning Spring Boot configurations for optimized performance.
- Utilizing Spring Boot Actuator for application insights and health checks.

Security with Spring Boot [1 Hour]

- Configuring Spring Security for authentication and authorization.
- Protecting endpoints and securing communication channels.

Container Orchestration [1 Hour]

- Introduction to Docker Compose for multi-container applications.
- Basics of Kubernetes for container orchestration.
- Deploying Spring Boot applications to a Kubernetes cluster.

Continuous Integration/Continuous Deployment (CI/CD) and Spring Boot [2 Hours]

- Automating Spring Boot application builds with CI/CD pipelines.
- Strategies for zero-downtime deployment in containerized environments.
- Logging and Monitoring



20 Hours

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